

A Study on Recycled Polymers Recovered from Multilayer Plastic Packaging Films by Solvent-Targeted Recovery and Precipitation (STRAP)

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Multilayer plastic film use increased in multiple packaging applications due to its versatility and overall increased performance over monolayer structures. However, the performance gains from multiple layers also make recycling difficult because they contain multiple polymers that can be immiscible and burdensome to traditional mechanical recycling operations. A possible solution is the solvent-targeted recovery and precipitation (STRAP) process, but the effect on the retrieved polymer is still unknown. The STRAP process is applied to two different multilayer films and samples of recovered polymers are evaluated for physical, molecular, and thermal properties. Changes in the molecular weight are not significant, but differences in thermal properties are reported along with the coprecipitation of different polymers. Solvent retention in the polymer matrix from STRAP reduced the glass transition temperature of samples, but enhanced drying recovered it. Heavy metals, such as Cd, Cr, and Pb are not detected, indicating regulatory compliance for different applications.

1. Introduction

Plastics are ubiquitous in everyone's lives, from cars to clothes or medical implants, due to their versatility and cost. Plastic production continues to grow with a global demand of 367 million tons in 2020^[1] and estimates of 460 million tons of waste plastic being generated per year by 2030.^[2] Considering large amounts of plastic are disposed in landfills or accumulating in the environment,^[3] plastic waste recycling became necessary for all

stakeholders focused on closed-loop recycling processes to divert recyclable streams from waste. The New Plastics Economy initiative, led by the Ellen MacArthur Foundation, is one example as consumer goods companies from multiple sectors, such as PepsiCo, Unilever, Danone, and Colgate-Palmolive, are committed to using recycled content in their plastic packaging up to 50 wt.% by 2025.^[4] In the United States (U.S.), recently approved bills in the states of California (Assembly Bill 793) and Washington (Senate Bill 5022) require increased recycled content in plastic beverage bottles up to 50% by 2030 and 2031, respectively.^[5] The European Union created a tax contribution of EUR 0.80 kg⁻¹ on nonrecycled plastic packaging waste, which became valid on January 1st, 2021.^[6]

Considering that packaging applications have the highest demand for plastic^[1] and the shortest product lifetime,^[3a] there is an immediate need to overcome the challenges associated with plastics recycling and increase the amount of recovered plastics. In the U.S. alone, it is estimated that packaging and food-service plastics, commonly single-use plastics (SUPs), comprise 16 million tons every year, equating to more than 45 kg per person.^[7] In addition to the food sector, the cosmetics and the pharmaceutical sectors also require sophisticated and complex packaging solutions that are difficult to recycle with current technologies.^[8] Multilayer flexible packaging films are usually comprised of multiple layers of different polymers, or aluminum foil, held together by polymeric tie layers. This gives the resulting packages specific functionalities to protect against contamination, enhance mechanical properties and chemical resistance, as well as increase the shelf life of the enclosed products.^[9] Some estimates state that 26% of the flexible packaging market is multilayer films.^[10] A pilot study carried out in the U.S. at a materials recovery facility (MRF) working with flexible plastic packaging determined that 10% of the output bale composition was composed of multilayer plastic structures.^[11]

Many polymers used in multilayer films are immiscible, such as polyethylene (PE) combined with polyamide (PA) or poly(ethylene terephthalate) (PET), which compromise the use of conventional and established mechanical recycling processes.^[12] Consequently, most multilayer flexible plastics are primarily sent to landfills or incinerated to produce energy.^[13] The accumulation of these structured materials is a result of both postcon-

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sumer recycled (PCR) plastic and postindustrial recycled (PIR) materials from production scraps and inefficiencies that could go up to 40% of the total production.^[14]

Due to the necessity of increasing the recycling rate of plastics and the difficulty in obtaining similar performance from single-layer or monolayer films, new methods are being explored to improve mechanical recycling and develop new selective and chemical recycling processes, which presents an increased potential for recycling heterogeneous plastic waste streams.^[15] A previous review from Kaiser et al.^[13] discussed some methods for recycling multilayer packaging, dividing them into two categories: processing all materials together with the use of a compatibilizer, such as the mechanical processing of PE/PA and PET/PE films, or the use of solvents to separate the different film layers.

The methods involving solvents are usually divided into those targeting the dissolution of binder layers for film separation or delamination of aluminum foil and those dissolving plastic layers with selective solvents and then recovering them in subsequent steps.^[16] Layer delamination was reported by Kulkarni et al.^[17] through the decomposition of plastic layers by sub- and supercritical water and Mahmood et al.^[18] through thermal disengagement. Another option is the dissolution of tie layers for recovery of the main polymer layers of a multilayer structure. For example, Ügdüler et al.^[9b] investigated the delamination of multiple multilayer films containing PET, PE, and polypropylene (PP), with polyurethane (PU) tie layers by utilizing carboxylic acids. Another study by Kaiser^[19] developed a thermoreversible cross-linked adhesive as a tie layer, which can be dissolved when heated with dimethylsulfoxide (DMSO).

The second solvent recycling method is based on the differences in polymer solubility, where the dissolution process is designed to solvate targeted layer(s) of the film independently or simultaneously. The dissolution is followed by polymer precipitation carried by solvent evaporation or introducing an antisolvent.^[20] One example is the solvent-targeted recovery and precipitation (STRP) process, a novel processing method based on the dissolution-precipitation framework to recover polymers and was first laid out by Walker et al.^[21] Specific solvents were used to dissolve each polymer for a PIR multilayer film containing PE, PET, and ethylene vinyl alcohol (EVOH) as main layers. The polymer solution is filtered followed by a precipitation step by cooling and/or by the addition of antisolvent. The process is repeated for the remaining unseparated layers but with a different solvent/antisolvent combination until all layers are independently recovered. The selection of the most suitable pair, its ratio, and the temperatures utilized for dissolution and precipitation was qualitatively predicted by a combination of methods, including the Hansen solubility parameters (HSP), molecular dynamics (MD) simulations, and quantum-based thermodynamic model, conductor-like screening model for real solvents (COSMO-RS). This approach was further refined and experimentally corroborated, as recently disclosed by Zhou et al.,^[22] for a more quantitative prediction of the temperature-dependent polymer solubilities for PE and EVOH by using MD and COSMO-RS models and employing a library with 524 solvents. Several companies are building pilot-scale facilities based on dissolution-precipitation processes, such as the CreaSolv Sachet Recycling plant in Indonesia by Unilever with a process developed by Fraunhofer Institute for Process Engineering and Packaging IVV

Table 1. Approximate polymer composition in films A1 and A2 used in the STRAP process.^[26]

Polymer	Film A1 (wt.%)	Film A2 (wt.%)
PE	4.00–4.50	3.50–4.50
EVOH	3.00–3.50	1.50–2.00
EVA	0.50–1.00	0.50–1.00
PET	92.50–93.50	72.50–73.50
PETG	Not present	20.00–21.00

and CreaCycle GmbH.^[23] Additional solvent-based methods for the recycling of multilayer films were recently reviewed by Kol et al.,^[24] including also solid-liquid extraction methods that can be used for the removal of unwanted compounds (e.g. additives) and deinking technologies (packaging label). The impacts on the viscosity of polymer solutions, produced as a result of solvent-based recycling processes, was also recently reviewed by Kol et al.,^[25] as the rheological behavior of these solutions can significantly affect some steps of the process, such as filtration efficiency.

In the current work, we characterized polymers recovered from the STRAP process based on the work of Sánchez-Rivera et al.^[26] The STRAP process was further optimized by Sánchez-Rivera et al. using more complex multilayer films and investigating the possibilities to reduce the use of antisolvents. Considering that most of the published literature on advanced plastic recycling methods focuses on new processes, this paper contributes to advancing the technologies via comprehensively characterizing the recovered polymers to assess their suitability for typical plastics applications. Polymer upcycling is the ultimate goal once unrecovered material can be reintroduced into material flows to achieve a closed-loop recycling process and a circularity model, which cannot currently be attainable only with mechanical recycling.

2. Experimental Section

2.1. Materials

Two different multilayer films supplied by Amcor (Neenah, WI) were subjected to the STRAP process, designated as films A1 and A2 described previously by Sánchez-Rivera et al.^[26] Film A1 is comprised of PET, PE, EVOH, and EVA (ethylene vinyl acetate). PE and PET were the most recovered polymers (>97 wt.%) and were the polymers characterized in this paper. Film A2 has glycol-modified poly(ethylene terephthalate) (PETG) in addition to the polymers recovered from film A1. The approximate composition of the films is displayed in **Table 1**. Two different precipitation techniques for film A1 were applied: one involving the use of antisolvent (STRAP-A) and a second one by solution cooling (STRAP-B). The technique used for film A2 used the cooling procedure for PE and EVOH separation and an antisolvent for PETG, PET, and EVA (STRAP-C). The details on the experimental procedure followed, including temperature, solvents, and time, were compiled in **Figure 1** for films A1 and A2, based on the work of Sánchez-Rivera et al.^[26] The exactly same virgin polymers used to manufacture the major film layers, PET, high-density PE (HDPE),

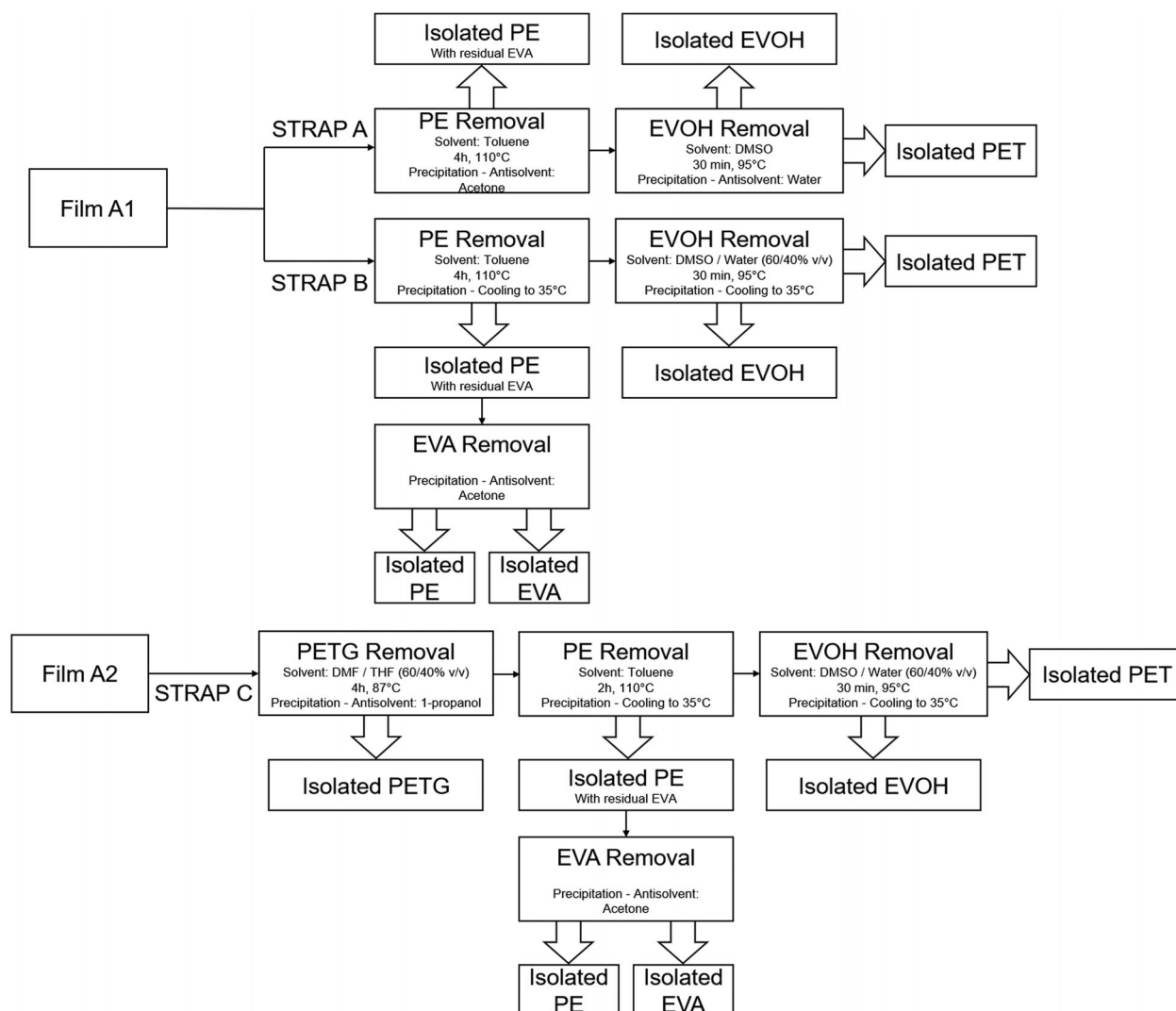


Figure 1. Simplified process schematic for the STRAP A, B, and C for films A1 and A2 based on the work of Sánchez-Rivera et al.^[26]

Table 2. Summary of experimental techniques utilized for the characterization of the recovered polymers and desired information.

Experimental technique	Desired information
Thermogravimetric analysis	Presence of volatile compounds as contaminants, thermal stability, inorganic residual
Headspace-gas chromatography-mass spectrometry	Identification of volatile compounds
Differential scanning calorimetry (DSC)	Thermal properties (melt temperature, glass transition temperature, crystallinity)
Melt flow rate (MFR)	Rheological properties (viscosity)
Gel permeation chromatography (GPC)	Molecular weight and molecular weight distribution
Fourier transformed infrared spectroscopy (FTIR)	Presence of contaminants (other polymers)
MicroRaman spectroscopy	Presence of contaminants (other polymers)
Inductively coupled plasma – optical emission spectroscopy (ICP-OES)	Presence and quantification of metals

and linear low-density PE (LLDPE) were characterized to compare properties of the recovered polymers and were in pellet form prior to film manufacturing. A summary with the characterization techniques used and the expected outcome is illustrated in **Table 2** at the end of this section.

2.2. Thermogravimetric Analysis

The thermal decomposition and total volatile content of the virgin and STRAP samples were determined via modulated thermogravimetric analysis (MTGA) using a Q5000IR thermogravi-

metric analyzer (TA Instruments, New Castle, DE). With a mass of 5–10 mg, the specimens were loaded to a platinum pan and heated at 2 °C min⁻¹ with continuous modulation using an amplitude = ± 5 °C and period = 200 s, under an N₂ atmosphere.

2.3. Headspace-Gas Chromatography-Mass Spectrometry

0.50 ± 0.01 g of each sample (STRAP and virgin resins) was placed into a headspace vial and sealed. The vials were heated in an oven for 10 minutes, at 180 °C for PET and PETG samples and 120 °C for PE samples. An air aliquot of 1.0 mL was collected from the vial and injected in an HP 6890 gas chromatograph (GC) system coupled to an Agilent 5973 mass selective detector (Agilent Technologies, Santa Clara, CA). An Agilent J&W DB-5 GC column (0.25 mm i.d., 0.25 µm) was used in split mode (2:1 ratio) with helium gas at 107.5 kPa. An initial equilibration time of 30 s at 40 °C was followed by a linear temperature increase to 270 °C at 12 °C min⁻¹ and then held at 270 °C for 10 min. Mass spectrometry ionization was carried by electron impact (70 eV) and the mass range between 50 and 550 m z⁻¹ was monitored.

2.4. Melt Flow Rate (MFR)

The melt flow rate for each sample was obtained using Procedure A of ASTM D1238-20 (ASTM, 2020a) with the specified parameters for polyethylene (190 °C, 2.16 kg) and poly(ethylene terephthalate) (285 °C, 2.16 kg) using a D4004 Melt Flow Indexer (Dynisco, Morgantown, PA). Samples were dried for 8 hours prior to the analysis at 70 °C and 140 °C, for PE and PET, respectively. PE samples were pre-heated for 5 min with a cut-off time of 1 min, for PET, the cutoff was 0.25 min. The measured mass was converted into grams per 10 min for all samples by the appropriate factor.

2.5. Gel Permeation Chromatography

The molecular weight and molecular weight distribution analysis of PE samples used 180–220 mg of polymer in HPLC grade 1,2,4-trichlorobenzene (TCB) (Fisher Scientific, Fair Lawn, NJ), containing 250 mg L⁻¹ of 2,6-di-tert-butyl-4-methylphenol (BHT) (Sigma Aldrich, St. Louis, MO) as an antioxidant. This system was left under constant agitation and heated to 150 °C for 5 h for complete dissolution resulting in a concentration of 10 mg mL⁻¹. The samples were injected at a volume of 200 µL using a Malvern Viscotek 350 HT-GPC (Malvern Panalytical, Westborough, MA), equipped with an internal filtration system and refractive index (RI), viscometer, and light scattering (LS) detectors. Chromatographic separation occurred using a flow rate of 1.0 mL/min and an oven temperature of 145 °C using two PLgel Olexis 300 × 7.5 mm columns (Agilent Technologies, Santa Clara, CA) in series with lower and higher molecular weight limits of 2 000 Da and 10 000 000 Da, respectively. A calibration curve was obtained using narrow polystyrene standards from 10 000 to 3 000 000 Da (Agilent Technologies, Santa Clara, CA) and converted for use with polyethylene using Mark-Houwink constants, as described

in ASTM D6474-20 (ASTM, 2020b) with the WinGPC software (PSS USA, Amherst, MA).

PET and PETG samples were prepared at a concentration of 5 mg mL⁻¹ by dissolving 6–20 mg of analyte overnight in HPLC grade hexafluoroisopropanol (HFIP) (Fisher Scientific, Fair Lawn, NJ), containing 1.7 g of sodium trifluoroacetate (Sigma Aldrich, St. Louis, MO) per kg of HFIP to improve signal quality. Samples were filtered through a 0.45 µm PTFE syringe filter before injection at a volume of 10 µL. The molecular weight distribution was obtained using an EcoSEC HLC-8320 GPC (Tosoh Bioscience, King of Prussia, PA) equipped with refractive index (RI) and ultraviolet (UV) detectors and with one TSKgel SuperH6000 150 × 6.0 mm column (Tosoh Bioscience, King of Prussia, PA) in series with two PL HFIPgel 250 × 4.6 mm columns (Agilent Technologies, Santa Clara, CA) to allow for better separation. A calibration curve was obtained using poly(methyl methacrylate) (PMMA) narrow standards from 2 000 to 1 600 000 Da (Agilent Technologies, Santa Clara, CA). The combination of columns used had a lower molecular weight limit of 200 Da and a higher molecular weight limit of 2 000 000 Da. The analysis was performed at a 0.3 mL min⁻¹ flow rate with the oven temperature at 45 °C.

2.6. Differential Scanning Calorimetry

The thermal transition properties for recovered polymers from the STRAP process were evaluated through a heat/cool/heat cycle between 10 and 180 °C for PE samples, and between 10 and 310 °C for PET and PETG samples, at a rate of 10 °C min⁻¹ for melt and crystallization (only PE and PET) and 20 °C min⁻¹ for glass transition (only PETG) under a N₂ atmosphere using a Q2000 differential scanning calorimeter (TA Instruments, New Castle, DE). Each specimen tested had a mass of 3–6 mg and was added into a hermetically sealed aluminum DSC pan and crimped prior to the analysis. The maximum possible polymer crystallinity was calculated using Equation (1) by dividing the heat of melting from the second heat cycle to remove any thermal history from previous film manufacturing processes by the reference heat of melting for a 100% crystalline polymer.

$$\text{Crystallinity} = \frac{H_{m,2}}{H_m^0} \times 100 \text{ (\%)} \quad (1)$$

2.7. Fourier Transform Infrared Spectroscopy

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) spectra were obtained using a Nicolet 6700 infrared spectrometer (Thermo Fisher, Waltham, MA) at ambient temperature (22 °C). A DTGS detector with 32 scans and a resolution of 2 cm⁻¹ was utilized and all spectra were analyzed with OMNIC™ 8.3 software (Thermo Fisher, Waltham, MA).

2.8. Raman Spectroscopy

Raman spectra were recorded on a DXR Raman Microscope (Thermo Fisher, Waltham, MA), collecting 32 scans with a 2 cm⁻¹

resolution, using a 780 nm laser excitation source, 60 mW laser power at the sample, and a 10x microscope objective. The results were analyzed with OMNIC 9.7 software (Thermo Fisher, Waltham, MA).

2.9. Inductively Coupled Plasma – Optical Emission Spectroscopy

Samples were evaluated for the presence of metals using inductively coupled plasma – optical emission spectroscopy (ICP-OES). Three replicates of 0.1500 ± 0.0005 g were collected for each blend analyzed and digested via microwave-assisted digestion using an UltraWave digestion system (Milestone, Inc., Shelton, CT) in 5 mL HNO₃ 67% v/v Trace Metal Grade (Fisher Scientific, Fair Lawn, NJ) and 1 mL HCl 34% v/v Trace Metal Grade (Fisher Scientific, Fair Lawn, NJ). An initial pressure of 40 bar with N₂ was applied. Microwave power of 1500 W was used for 40 min, with a temperature increase from room temperature (22 °C) to 260 °C and a system pressure increase to 150 bar, followed by a subsequent cooling period of 20 min. The method employed is a slight modification of the modified Westerhoff digestion method described by Goodlaxson et al. (2018).

The digested samples were diluted to 50 mL with ultra-pure, deionized water and analyzed using an iCap-7400 Duo ICP-OES (Thermo Scientific, Waltham, MA). The wavelength with the lowest limit of detection (LOD) was used for each metal analyzed, with all measurements done in radial mode. The multi-element standard contained 0.1, 1, 5, 25, 50, and 100 µg/mL of each metal of interest, along with a 5 µg mL⁻¹ yttrium internal standard. All standard solutions were prepared from single standard solutions of 1000 µg mL⁻¹ (Inorganic Ventures, Christiansburg, VA). The metals tested were Al (aluminum), Cd (cadmium), Cr (chromium), Fe (iron), Pb (lead), Sb (antimony), and Ti (titanium). Samples were run with concurrent blanks (no polymer added) for each digestion batch in order to discard incidental metals present in the acids or leached from the vessels used in the digestion.

2.10. Statistical Analysis

The results herein presented were analyzed for the difference between means using one-way ANOVA considering a 95% confidence level ($\alpha = 0.05$) and grouped using Tukey's honestly significant difference (HSD) test. JMP 15 Pro (SAS Institute Inc., Cary, NC) was utilized as the statistical analysis software. All analyses utilized five repeated measures unless otherwise noted.

3. Results and Discussion

3.1. Influence of STRAP Process on Recovered Polymer Properties

3.1.1. Melt Flow Rate

The melt flow rate (MFR) was used to determine the impact of the STRAP process on the viscosity of the PE and PET samples. The analysis determined that PE STRAP samples possessed a

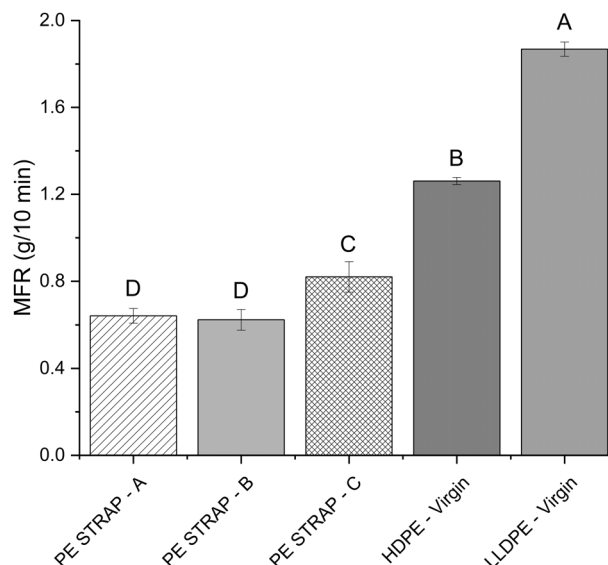


Figure 2. Melt Flow Rate of PE samples recovered via STRAP and virgin resins used in the multilayer films. Data points associated with the same letter are statistically equivalent ($\alpha = 0.05$).

lower MFR (higher viscosity) when compared to the virgin HDPE and LLDPE (Figure 2). PE STRAP A and B were not statistically different ($p > 0.05$) whereas PE STRAP C possessed a higher MFR (low viscosity) than STRAP A and STRAP B recovered polymers, although still statistically lower than the virgin resins. A possible cause for this result can be associated with polymer contamination (e.g. PET), as observed with the Raman analysis (see cross-contamination between polymers sections) and the presence of solvents, determined with the thermogravimetric analysis (see solvent entrapment on polymer matrix). A lower MFR value would suggest the presence of crosslinking, however, there was no increase in molecular weight as observed with the gel permeation chromatography results (see Molecular weight determination subsection), which support that polymer contamination is the probable cause for MFR reduction in PE STRAP samples. Additional oscillatory rheology experiments would help for a better understanding of degradation mechanisms during the polymer recovery process. PET STRAP samples presented values with greater variability, with A, B, and C samples not being statistically different between themselves, only from the virgin PET, which had a significantly lower MFR value, as shown in Figure 3. We hypothesize that the reduction in viscosity observed in STRAP PET can be attributed to the presence of residual solvents, such as the DMSO and toluene entrapped in the polymer matrix observed from TGA and HS-GC-MS analyses (discussed later), which increases chain mobility and chain rearrangement. Trace amounts of polyethylene in PET were also observed via Raman spectroscopy. Also, the PET processing from pellets to film, during the multilayer film manufacturing, has the potential to reduce the viscosity of the polymer due to hydrolysis.

3.1.2. Molecular Weight Determination

The molecular weight and molecular weight distribution of each STRAP recovered polymer was assessed to understand the im-

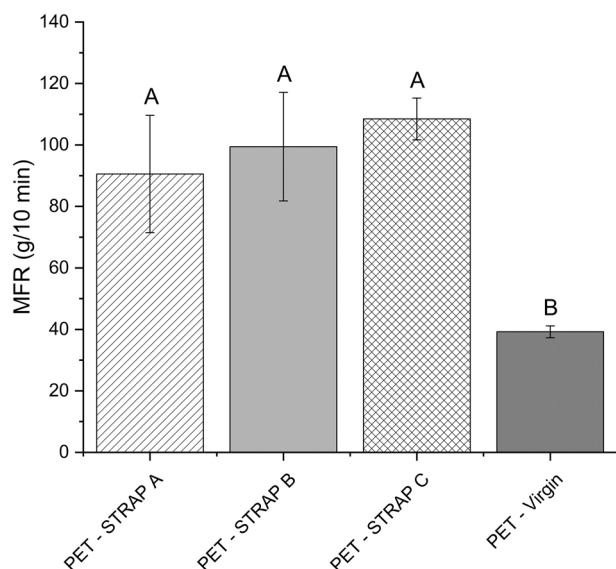


Figure 3. Melt Flow Rate of PET samples recovered via STRAP and virgin resins used in the multilayer films. Data points associated with the same letter are statistically equivalent ($\alpha = 0.05$).

part of the STRAP process on polymer degradation. The polyethylene samples recovered from the STRAP process possessed lower number-average (Mn) and weight-average (Mw) molecular weight compared to both virgin HDPE and LLDPE resins possibly indicating some chain scission occurred during the film manufacturing process, as a result of processing conditions (e.g. temperature) or during the dissolution-precipitation process. We hypothesize that the film production is a more probable cause for the reduction in the molecular weight due to its nature as a physical process compared to STRAP, a chemical process. Additional testing for each single layer of the film prior to STRAP would be required to identify which process caused the observed changes. Unfortunately, these samples were not provided by the film supplier. The dispersity (D) of all PE STRAP samples was lower (D between 4.43 and 5.47) than the virgin HDPE (bimodal distribution, $D = 5.97$), possibly suggesting that the presence of LLDPE (unimodal distribution, $D = 3.00$) was one of the factors responsible for the reduction observed. Also, possible contamination with EVOH was possible, although its molecular weight was not tested. Benoit et al.^[27] observed that the Mw of HDPE was reduced with the increase of mechanical recycling cycles, with Mn remaining almost constant leading to a reduction in dispersity. It was suggested that chain scission was the main cause for the result, with mostly longer chains being reduced to medium chains. Oblak et al.^[28] also observed a reduction in Mn and Mw with HDPE mechanical recycled up to 100 cycles, suggesting the effect of chain scission on the molecular weight difference. We hypothesize that a similar mechanism occurred with the PE STRAP samples, as the reduction in Mw was more pronounced than the Mn (Table 1), but some chain scission also occurred in smaller polymer chains due to the reduction in the Mn values.

Previous studies for PET have demonstrated that thermo-mechanical recycling results in a decline in performance properties which impacts its potential use as a recycled resin.^[29] The comparison between virgin and STRAP samples shows that

Table 3. Molecular weight and dispersity for virgin and STRAP samples.

Resin	STRAP	Mn [g mol ⁻¹]	Mw [g mol ⁻¹]	D
PE	A	21,540	117,910	5.47
PE	B	21,300	94,420	4.43
PE	C	16,410	88,760	5.41
HDPE	Virgin	27,290	162,920	5.97
LLDPE	Virgin	34,370	103,090	3.00
PET	A	12,200	37,100	2.36
PET	B	12,600	28,900	2.26
PET	C	13,500	28,500	2.29
PET	Virgin	14,400	30,900	2.57
PETG	C	12,000	25,500	2.12
PETG	Virgin	12,000	23,000	1.93

Mn: Number Average Molecular Weight; Mw: Weight Average Molecular Weight; D: Dispersity.

the recovered samples (D between 2.29 and 2.36) present a slightly narrower molecular weight distribution when compared to the virgin resin ($D = 2.57$), though the Mn was not practically different. Similar results were noticed by Poulakis and Papaspyrides^[30] using a different solvent/antisolvent system suggesting that the STRAP process influenced these properties. Using a different system, Goje^[31] noticed only a slight decrease in viscosity-average molecular weight of recycled PET by the dissolution/precipitation technique. As illustrated in Table 3, the Mw was slightly higher for PET STRAP A, with STRAP B and C showing lower values and closer to the one observed for the virgin PET sample. A possible explanation could be an increased amount of contamination with PE, which showed a higher Mw than PET samples. The GPC results offer some evidence that the increase in the MFR of PET STRAP samples is not associated with degradation of the molecular chain and can corroborate the hypothesis of entrapped solvent or contamination with PE/EVOH that degrade at the higher temperatures used at PET analysis/processing. The latter can be linked to the significant difference in viscosity observed in MFR, which uses a more sample material for testing, and, thus more susceptible to containing contaminants when compared to the GPC analysis. The comparison between virgin and STRAP PETG showed a similar trend as observed for PET STRAP B and C, with smaller Mw and narrower dispersity ($D = 2.12$ vs. $D = 1.93$, respectively) for the solvent-recovered sample.

Polymer crystallinity was investigated for PE and PET samples using the second heat of the DSC experiments (Figures 4 and 5). The second heat cycle was used to understand the crystalline structures from the melt under controlled conditions and enables an equal comparison among samples without interferences from the previous heat history. It was observed that PE STRAP samples possessed a crystalline fraction between the virgin HDPE and LLDPE resins used in the film manufacturing, with values closer to HDPE, which suggests that it is a mixture of the resins, possibly following the law of mixtures observed with other polymer properties in the mixtures of different density polyethylene.^[32] While there was no significant difference ($p > 0.05$) between PE samples processed from STRAP B and C, the

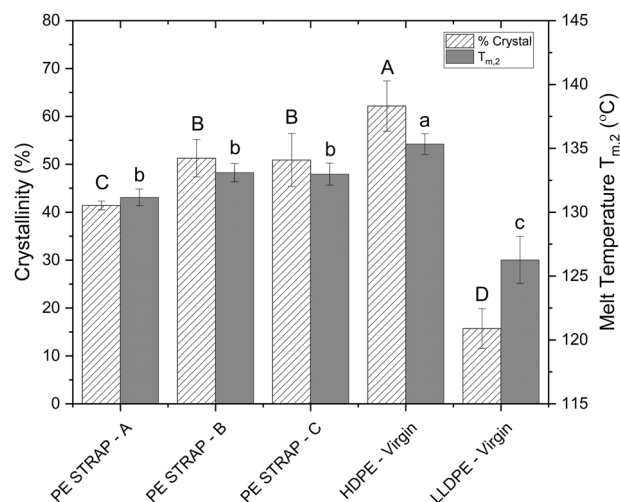


Figure 4. Crystallinity and melt temperature (second heat cycle) of PE samples recovered via STRAP and virgin resins used in the multilayer films. Data points associated with the same letter (uppercase for crystallinity and lowercase for temperature) are statistically equivalent ($\alpha = 0.05$).

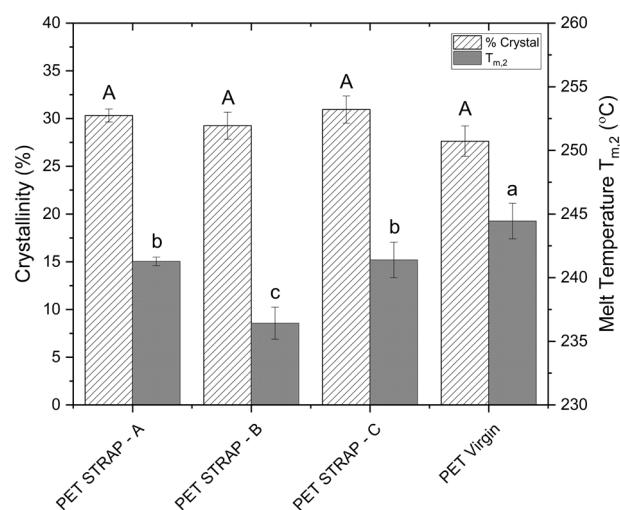


Figure 5. Crystallinity and melt temperature (second heat cycle) of PET samples recovered via STRAP and virgin resins used in the multilayer films. Data points associated with the same letter (uppercase for crystallinity and lowercase for temperature) are statistically equivalent ($\alpha = 0.05$).

crystallinity between STRAP A and B samples is significantly different ($p < 0.05$). The process using antisolvent possessed $\approx 10\%$ lower crystallinity than the process using the cooling down the procedure for polymer precipitation. Another hypothesis for the reduced crystallinity could be associated with residual EVA increasing the amount of amorphous region, not specifically removed in STRAP A but in STRAP B.

The crystallinity of PET samples was not statistically different between the virgin resin and STRAP-recovered samples. This result suggests that toluene and DMSO fractions found in the samples did not hinder the crystallinity potential due to creating strong polymer-solvent interaction as the solvent promotes a structural rearrangement and swelling and in turn, favors polymer crystallinity, with the effects of each solvent based on the

Table 4. Thermal properties collected from the second heat cycle of DSC analysis.

Resin	STRAP	$T_{m,2}$ (°C)	$H_{m,2}$ [J g ⁻¹]	%Crystallinity	T_g (°C) ND	T_g (°C) D
PE	A	131.2 ^a	121.3 ^g	41.4 ^l		
PE	B	133.1 ^a	150.2 ^h	51.3 ^m		
PE	C	133.0 ^a	149.1 ^h	50.9 ^m		
HDPE	Virgin	135.3 ^b	182.2 ⁱ	62.2 ⁿ		
LLDPE	Virgin	126.3 ^c	46.0 ^j	15.7 ^o		
PET	A	241.3 ^d	42.5 ^{j,k}	30.3 ^{p,q}	63.4 ^r	80.1 ^u
PET	B	236.4 ^e	41.0 ^{j,k}	29.3 ^{p,q}	56.6 ^s	80.5 ^u
PET	C	241.4 ^d	43.3 ^{j,k}	30.9 ^p	60.5 ^{r,s}	80.3 ^u
PET	Virgin	244.5 ^f	38.7 ^k	27.6 ^q	81.6 ^t	-

D: dried samples at 180 °C for 8 h, ND: samples not dried. Values associated with the same letter are statistically equivalent ($\alpha = 0.05$).

Flory-Huggins interaction parameters.^[33] A reduction in T_g of the STRAP PET was determined (≈ 20 °C) due to the presence of solvents (confirmed by HS-GC-MS and TGA) absorbed by the polymer having a plasticizing effect,^[34] lowering the activation energy for segmental motion.^[35] These observations support the measured increased MFR values observed.

The analysis of the second heat cycle ($T_{m,2}$) melt temperature showed that PE STRAP samples presented values between the ones observed for virgin HDPE and LLDPE resins (Figure 3). The differences between processes A, B, and C were not statistically significant, which could suggest a statistical mixture of the PE resins in the film structure according to the law of mixtures.^[32] However, for the PET samples (Figure 4), all STRAP materials possessed lower melt temperatures than the virgin resin, with processes A and C being statistically equivalent (241.3 °C and 241.4 °C, respectively) and STRAP B presenting the lowest value (236.4 °C) of all resins tested. The presence of solvents entrapped on the PET matrix is also a factor that lowers the melt temperature, although not significantly as observed with the glass transition temperature (T_g).^[34] This result confirms that PET STRAP B had a higher amount of solvent when compared to STRAP A and C as a lower T_g was observed, agreeing with TGA results displaying higher mass loss near 180 °C. (Figure 6 – PET TGA). After drying the samples at 180 °C for 8 h, they were reanalyzed by DSC, and it was noticed that the T_g for all PET STRAP samples increased (Table 4) to similar values as the virgin PET. Figure S1, Supporting Information, shows a representative thermogram of PET STRAP B samples before and after drying at 180 °C for 8 h. This result indicates a possible solution for removing solvents entrapped in the PET samples, although optimization studies are required to reduce energy consumption and production time and reduce the degradation of PE/EVOH contaminants that were not completely removed during the STRAP process.

3.2. Solvent Entrapment on the Polymer Matrix

3.2.1. Thermogravimetric Analysis

Thermogravimetric analysis was used to determine the presence of residual solvent used in the STRAP process within the polymer matrix. Recovered polyethylene samples clearly contained

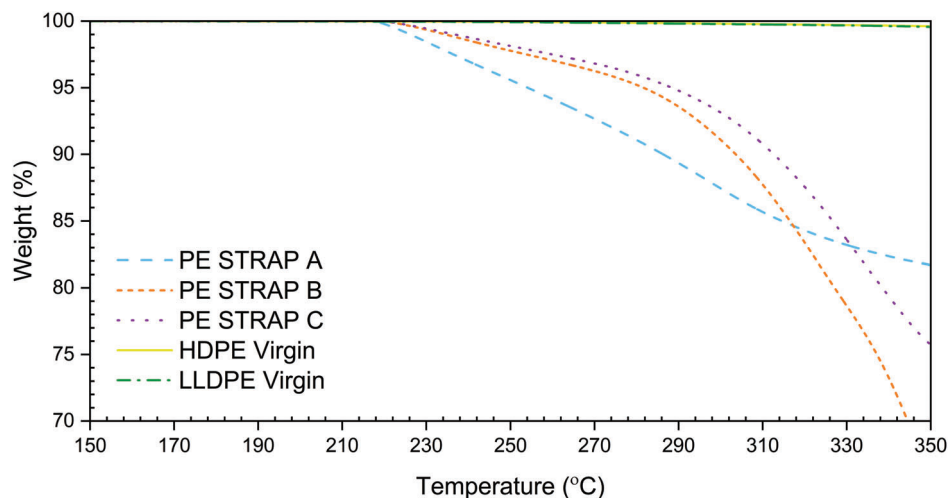


Figure 6. Decomposition profile between 150 and 350 °C of STRAP and virgin PE sample.

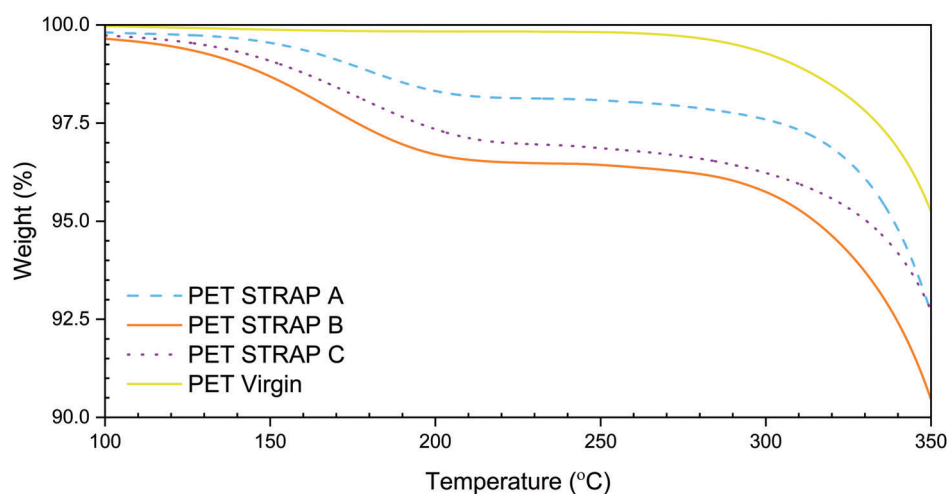


Figure 7. Decomposition profile between 100 and 350 °C of STRAP and virgin PET sample.

higher total volatile compared to the virgin samples (Figure 6), with the former presenting significant mass loss at early temperatures (≈ 210 °C) when compared to the virgin HDPE and LLDPE (≈ 400 °C). One possible cause could be associated with the presence of EVA, especially for PE STRAP A (EVA is not removed from PE), as a shoulder is observed between 350–400 °C, is expected from acetic acid evolution (from acetate groups).^[36] The presence of solvents can also lower the temperature of the first mass loss event, as further explored with HS-GC-MS. For STRAP PET samples, there is evidence of substances volatilizing at lower temperatures than the virgin PET, as illustrated in **Figure 7** by the mass loss events around 160–200 °C, when mass loss accounted for between 2 and 3.5 wt.%. For virgin resin, degradation only started around 275 °C.

3.2.2. Headspace-Gas Chromatography-Mass Spectrometry

Identification of solvents present in the STRAP samples was qualitatively conducted using headspace-gas chromatography-

mass spectrometry (HS-GC-MS). The presence of dimethyl sulfoxide (DMSO, boiling point = 189 °C) was detected in all PET STRAP samples, with a retention time of ≈ 5.7 min (**Figure 8**) for PET STRAP B with other samples included in the supplementary information. Toluene (boiling point = 111 °C) was detected for PET STRAP B and C, with a retention time of ≈ 3.7 min. Both mass spectra from the chromatogram peaks possessed highly correlated mass spectra ($>90\%$) with known spectra in the NIST Mass Spectra Library. These specific solvents are utilized in the STRAP process for selective solvation of polymer layers, according to Sánchez-Rivera et al.^[26] Specifically, DMSO is used as the EVOH selective solvent, whereas toluene is used as a selective solvent for PE. The lack of toluene in PET STRAP A (**Figure S2**, Supporting Information) could be associated with the use of water in the EVOH selective solvent (60% v/v DMSO/40% v/v water) for STRAP B and C, while STRAP A uses only DMSO. For PET STRAP C (**Figure S3**, Supporting Information), dimethylformamide (DMF, boiling point = 153 °C) and tetrahydrofuran (THF, boiling point = 66 °C) used as the PETG selective sol-

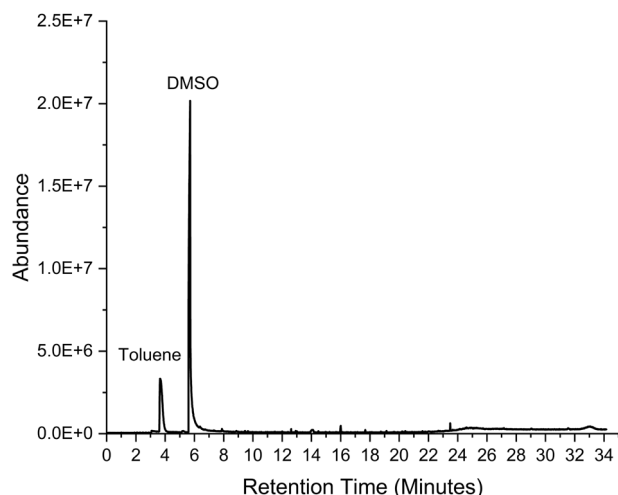


Figure 8. Chromatogram profile for PET STRAP B displaying peaks associated with the mass spectrum of toluene and DMSO.

vents were not detected in the analysis, although the peak associated with toluene presented similar intensity as the DMSO peak, which differentiates from PET STRAP B where it was smaller.

For PE STRAP samples, only PE STRAP C (Figure S4, Supporting Information) contained residual toluene and DMF associated with the PETG selective solvent, which could explain the mass loss at lower temperatures observed with the TGA analysis. However, PE STRAP B showed similar behavior and toluene was not detected, which was the only solvent expected to be found for PE STRAP A and B. Solvents were also not detected on PETG STRAP samples, and the virgin PE, PET, and PETG samples analyzed, indicating that they result from the STRAP process.

3.3. Cross-Contamination between Polymers

3.3.1. Fourier-Transformed Infrared analysis

The ATR-FTIR spectra of polyethylene samples (**Figure 9**) were normalized at the methylene band of 1368 cm^{-1} , commonly used as a reference band and for the carbonyl index calculation^[37] to facilitate comparison between samples. The presence of some peaks in the STRAP recovered PE was associated with characteristic bands of ethylene vinyl acetate (EVA) and EVOH polymers also present as film layers in the original material or from PET and PETG (for STRAP-C). The carbonyl ($\text{C}=\text{O}$) stretching at 1740 cm^{-1} ,^[38] which is detected in EVA as an ester bond from the acetate group, is present or from reminiscent PET embedded into the PE and not recovered by the STRAP process. According to Smith,^[39] another prominent peak located at 1240 cm^{-1} is commonly assigned to $\text{C}-\text{O}$ stretching ($\text{C}-\text{C}-\text{O}$ stretch) and is unique to acetates. The HDPE and LLDPE used with the films presented a peak around 1045 cm^{-1} not characteristic of PE, which can be attributed to an additive with oxygen since this peak is usually assigned to $\text{O}-\text{C}-\text{C}$ stretching in esters.^[39] The STRAP PE presents a shifted peak at 1020 cm^{-1} , which we hypothesize is associated with possible solvation from residual solvents in the polymer matrix, as frequency shifts are common to changes in the local environment by electrostatic interactions.^[40]

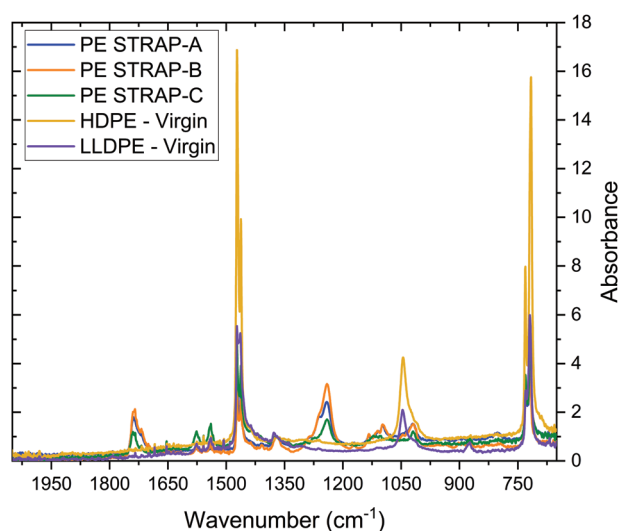


Figure 9. ATR-FTIR spectra of PE samples recovered via STRAP and virgin resins used in the multilayer films.

The spectra for the PET samples shown in Figure S4, Supporting Information, (normalized at 1410 cm^{-1} , using the benzene ring band as the reference band^[41]) present the most significant differences between virgin and STRAP samples at the wavelengths associated with $\text{C}-\text{H}$ stretching vibrations around 2916 and 2848 cm^{-1} and $\text{C}-\text{H}$ bending vibration at 1462 cm^{-1} , both strong vibrations characteristic of PE.^[42] It suggests the presence of PE on the STRAP PET samples, especially for STRAP A and C samples. All ATR-FTIR peak assignments for PE, PET, PETG, and EVA are compiled in Table S1, Supporting Information.

3.3.2. MicroRaman Analysis

Two peaks present in the PE STRAP C Raman spectra (Figure S6, Supporting Information; normalized at 1368 cm^{-1}) are not associated with PE but with PET. The peak at 1730 cm^{-1} refers to the carbonyl stretching, while the peak at 1615 cm^{-1} is associated with $\text{C}-\text{C}$ ring stretching and the $\text{C}-\text{H}$ ring bending; both vibrations are associated with polymers containing an aromatic ring.^[43] These observations suggest the contamination of PE STRAP-C with PET was also noted with the FTIR spectra. Small white spots on the surface of PE STRAP-C were investigated using MicroRaman imaging. The Raman spectra of these spots are closely associated with PET (**Figure 10a**) providing more definitive evidence of PET contamination. White spots were visible on the STRAP PET samples with characteristic peaks of PE as illustrated in Figure 10b for the PET STRAP-B; otherwise, no significant difference was observed. All Raman peak assignments for PE, PET, PETG, and EVA are compiled in Table S2, Supporting Information.

3.3.3. Inorganic Contamination: Metal Content Analysis

The metal content was determined in the virgin and recovered samples through the STRAP process using ICP-OES. None of

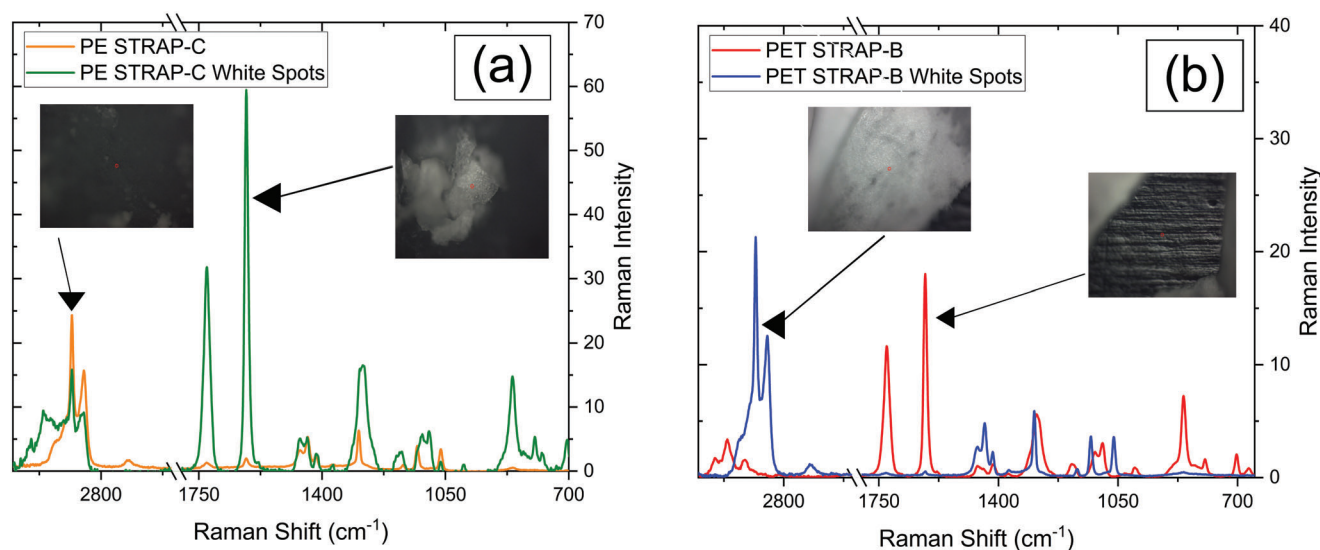


Figure 10. Raman spectra of PE a) and PET b) samples recovered via STRAP containing white spots.

Table 5. Metal composition of selected compounds (Al, Sb, Fe, Ti) in the original films, virgin resins, and STRAP samples.

Resin	STRAP	Al (ppm)	Sb (ppm)	Fe (ppm)	Ti (ppm)
PE	A	178.04	24.220	7.8309	0.6061
PE	B	187.25	15.462	15.382	1.8941
PE	C	208.08	a)	16.754	7.0243
HDPE	Virgin	98.450	b)	a)	a)
LLDPE	Virgin	105.12	b)	a)	2.1380
PET	A	88.091	181.35	b)	b)
PET	B	132.76	178.91	a)	b)
PET	C	114.35	198.88	b)	b)
PET	Virgin	85.598	129.61	a)	2.7100
PETG	C	177.46	a)	2.3287	13.230
PETG	Virgin	166.85	b)	b)	8.3756
Film A1	-	100.06	170.16	b)	b)
Film A2	-	76.646	149.57	a)	2.3116
Method LOD (ppm)		0.0143	0.0127	0.0038	0.0008
Method LOQ (ppm)		0.0477	0.0423	0.0127	0.0027

a) Value between LOD and LOQ; b) Value below LOD.

the samples analyzed had detectable concentrations of Cd, Cr, Pb, and Ni, which is necessary for compliance with the Toxics in Packaging Legislation, framework legislation in effect in 19 states of the United States.^[44] This legislation states the maximum of 100 ppm of the sum of Cd, Cr⁺⁶ (hexavalent chromium), Pb, and Hg (mercury, not tested in the work but historically not detected). Similar legislation is also in effect in Europe, according to Article 11 of the European Parliament and Council Directive 94/62/E.C.^[45]

Al, Sb, Fe, and Ti were detected in the analyzed samples (Table 5), as these metals are commonly associated with polymerization catalysts. Antimony (Sb) is a common catalyst for PET

synthesis^[46] and was detected in all PET samples below 200 ppm. Aluminum, iron, and titanium are metals present in Ziegler-Natta and metallocene catalysts used for polyethylene synthesis and were expected to be found since Cr was not observed in the samples, another metal associated with Phillips catalysts also used in PE manufacturing.

4. Conclusion

This study comprehensively characterized polyethylene (PE) and poly(ethylene terephthalate) (PET) recovered through the use of the STRAP process from two different multilayer films. This study determined that using different dissolution solvents and processing temperatures (STRAP A, B, C) influences the measured physical properties of STRAP recovered polymers. We show in this work that both PET and PE recovered through the STRAP process possess different viscosity parameters when compared to the virgin resins used in film manufacturing. This result can be associated with the presence of contamination from other polymers present in the film structure, with only a slight reduction in molecular weight observed for STRAP samples, possibly suggesting chain scission as a cause for the degradation. The contamination was confirmed using ATR-FTIR and Raman spectroscopy, including the presence of EVA and EVOH used as tie layers in the multilayer film in PE and PET recovered resins.

The thermal properties analyzed showed differences in melt temperature and crystallinity associated with the presence of solvents (mainly DMSO and toluene used in the dissolution step) entrapped in the polymer matrix, as shown in the headspace-GC-MS and TGA analysis. It was demonstrated that solvent removal could be achieved by employing a higher drying temperature, as there was a recovery in glass transition temperatures (T_g) measured for the PET resin. The inorganic analysis did not show the presence of Cd, Cr, and Pb, regulated metals for food packaging materials in the United States, which can suggest the safety of the STRAP-recovered plastic resins. Future studies

are needed to include mechanical testing, which will require the production of a more significant amount of STRAP resins for the injection molding of the samples to better understand product applications for STRAP resins. Additional, analysis with PCR polymers collected from material recovery facilities (MRFs) and improvements to the STRAP process are needed to reduce polymer contamination and reduce solvent entrapment, especially for PET, although it was demonstrated that improved drying conditions are a possible solution.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

advanced recycling, flexible plastics recycling, multilayer plastic films, recovered plastics characterization, solvent-targeted recovery and precipitation recycling

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